

# SPATIAL AND SPATIO-TEMPORAL MODELS WITH INLABRU

Girona, 6-8 May 2026

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## 1 Welcome to the course!

- Welcome to the `inlabru` workshop in Girona!
- The aim of this workshop is to introduce you to a range of statistical modelling approaches, in particular the temporal, spatial and spatio-temporal modelling as implemented in the `inlabru` package.
- Workshop materials are available in the github repository

 Important! □

Please use the instructions provided [here](#) to install and check your installation *before* the course starts.

## 2 Learning Objectives for the workshop

At the end of the workshop, participants will have an understanding of:

- the motivation for and the challenges of analysing and modelling spatial data
- statistical models used to analyse spatial and spatio-temporal data
- the implementation of these models in the `inlabru` package
- how to independently analyse spatial data with `inlabru`

## 3 Intended audience

The workshop aims to cater for participants with a range of different backgrounds, who are interested in analysing data with modern spatial and spatio-temporal statistical modelling approaches.

## 4 Prerequisites

Participants should be familiar with the R environment, and general statistical approaches for modelling such as regression, analysis of (co)variance, and generalized linear models.

No knowledge of R-INLA or `inlabru` is required.

## 5 Schedule

**NOTE** The schedule is indicative and might change as we work through the material.

### 5.0.1 Day 1

| Time          | Topic                                             |
|---------------|---------------------------------------------------|
| 10:00 - 10:30 | <b>Welcome and intro to the course</b>            |
| 10:30 - 11:30 | <b>Session 1:</b> Introduction to inlabru         |
| 11:30 - 13:00 | <b>Session 2:</b> Some practical tips on inlabru  |
| 13:00 - 14:30 | Lunch break ☐                                     |
| 14:30 - 15:30 | Practical Session 1                               |
|               | Practical Session 2                               |
| 15:30 - 16:30 | <b>Session 3:</b> Time Series Models              |
| 16:30         | <b>Session 3a:</b> comparison INLA versus inlabru |

### 5.0.2 Day 2

| Time          | Topic                                                |
|---------------|------------------------------------------------------|
| 9:15 - 11:00  | Practical Session 3                                  |
| 11:00 - 12:00 | <b>Session 4:</b> Introduction to Spatial Statistics |
| 12:00 - 13:00 | Lunch break ☐                                        |
| 13:00 - 14:00 | Practical Session 4                                  |
| 14.00 - 15:00 | <b>Session 5:</b> Areal processes                    |
| 15:00 - 16:00 | <b>Session 6:</b> Geostatistics                      |

### 5.0.3 Day 3

| Time          | Topic                                                      |
|---------------|------------------------------------------------------------|
| 09:15 - 11:00 | Practical Session 5                                        |
| 11:00 - 12:00 | <b>Session 7:</b> Point Processes                          |
| 12:00 - 13:00 | Lunch break ☐                                              |
| 13:00 - 14:00 | Practical Session 5 (continued)                            |
| 14:00 - 15:00 | <b>Session 8:</b> Spatiotemporal models                    |
| 15:00 - 16:00 | <b>Session 9:</b> Multi-likelihood/joint likelihood models |
| 16:00 - 17:00 | Closing session and discussion                             |

## 6 Licence

CC-BY-4.0

## 7 Citation

Belmont J., Martino S., Illian, J. F (2026). *inlabruCourseMarch2026: Course Materials for Noumea inlabru Course 10-14 March 2026*. <https://inlabru-courses.github.io/NewCaledonia2026/>

```
@Manual{Belmont2026inlabru,
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  author = {Belmont, Jafet and Martino, Sara and Illian, Janine},
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year = {2026},  
url = {https://janineillian.github.io/Girona2026/index.html}  
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